Check for updates

WILEY

Journal of Computer Assisted Learning

ORIGINAL ARTICLE

Exploring ChatGPT-Facilitated Scaffolding in Undergraduates' Mathematical Problem Solving

Ruijie Zhou[1,2] | Xiuling He[1,2] | Qiong Fan[3] | Yangyang Li[1,2] | Yue Li[1,2] | Xiong Xiao[1,2] | Jing Fang[1,2]

[1]National Engineering Research Center of Educational Big Data, Central China Normal University, Wuhan, China | [2]National Engineering Research Center for E-Learning, Central China Normal University, Wuhan, China | [3]School of Mathematics and Statistics, Central China Normal University, Wuhan, China

Correspondence: Xiuling He (xlhe@ccnu.edu.cn) | Qiong Fan (xfanqiong@ccnu.edu.cn)

Received: 25 September 2024 | **Revised:** 30 April 2025 | **Accepted:** 28 May 2025

Funding: This work was supported by the National Natural Science Foundation of China (NSFC) for the Project 'A Study on the Perception and Attribution Analysis of Learners' Higher-Order Thinking Activities' (62177023). This work is conducted as part of the 'AI + Evaluation' Special Teaching Reform Project of Central China Normal University for the Project 'AI+ Data Empowered Online Course Learning Quality Evaluation Research' (CCNU24JG11) and the Excellent Graduate Education Innovation Funding Project of Central China Normal University for the Project 'Generative Artificial Intelligence Enabled Programming Model Innovation and Practice Research for College Students' (2024CXZZ059).

Keywords: behavioural analysis | ChatGPT | learning scaffolding | mathematical problem solving | perception

ABSTRACT

Background: ChatGPT, an AI-based chatbot, supports learning by accurately interpreting and responding to user inputs. Despite its potential, few empirical studies have examined its influence on college students' mathematical problem-solving processes.

Objectives: This study aimed to introduce a ChatGPT-facilitated scaffolding to investigate its impact on students' mathematical problem-solving behaviours, performance and perceptions.

Methods: Twenty-nine undergraduates participated in this study, engaging in mathematical problem-solving tasks using the scaffolding. A mixed-method approach was employed, incorporating performance data, interaction analysis and self-reported surveys to assess both quantitative and qualitative aspects of students' experiences. In particular, lag sequential analysis was applied to explore the undergraduates' problem-solving behavioural patterns.

Results and Conclusions: Results demonstrated that the ChatGPT-facilitated scaffolding significantly improved students' mathematical problem-solving performance. The high-performance group (HPG) exhibited a greater frequency of interpretive and evaluative activities, transitioning from factual to metacognitive representations, while the low-performance group (LPG) primarily transitioned from prompt selection to procedural representations. Additionally, most participants expressed positive perceptions of the scaffolding experience and reported an improvement in their problem-solving skills.

Major Takeaways: These findings offer valuable insights for the design and implementation of AI-facilitated learning activities in mathematical problem-solving contexts.

1 | Introduction

Problem solving is a critical element of the mathematics curriculum and a key indicator of learning success (Blum and Niss 1991; Säfström et al. 2024). This process requires students to integrate their knowledge and skills to construct essential components of the solution (Csanadi et al. 2021). However, due to the diverse mathematical abilities among students, they encounter various challenges when addressing tasks, which can adversely affect their overall educational experience (Gabriel et al. 2020; Jonte et al. 2022). Moreover, flexibility is essential in the problem-solving process, necessitating

© 2025 John Wiley & Sons Ltd.

Summary

- What is already known about this topic
  - Providing personalised mentoring in mathematics remains a significant challenge.
  - ChatGPT offers potential support for learners during mathematical problem solving.
  - Few studies have examined how students engage in problem solving with ChatGPT.
- What this paper adds
  - ChatGPT-facilitated scaffolding is proposed to support mathematical problem solving.
  - Improvements were observed in both student performance and perceptions.
  - Distinct problem-solving behaviours emerged across different performance groups.
- Implications for practice and/or policy
  - The findings reference the design and implementation of AI learning activities.
  - The scaffolding connects mathematical problem-solving processes with ChatGPT.

that students adapt strategies effectively to solve problems (National Research Council and Mathematics Learning Study Committee 2001). Despite these requirements, the limited availability of time and resources makes it difficult for educators to provide personalised mentoring (Høynes et al. 2019). Consequently, there is a compelling need to provide robust support to college students in their mathematical problem-solving endeavours.

To enhance student problem-solving efficacy, a variety of technological strategies have been examined (Kim and Hannafin 2011; Lu and Xie 2024). Scaffolding, in particular, has garnered significant attention in education due to its ability to help students acquire knowledge and skills gradually throughout the learning process. Research has increasingly focused on distinct problem-solving stages, improving cognitive outcomes through tailored scaffolding in conceptual, metacognitive, procedural and strategic domains based on specific research goals and contexts (Ahmadi Safa and Motaghi 2024; Kim and Lim 2019; Kim and Hannafin 2011). Scaffolding in mathematical problem-solving is typically diverse and adaptable, designed to meet varied educational needs (Álvarez et al. 2020; Marco and Palatnik 2024). Studies have integrated different types of scaffolding to comprehensively support various stages of problem-solving and to tailor interventions to student needs (Lange et al. 2023; Ouyang et al. 2021). However, these adaptable strategies have predominantly relied on rule-based algorithms, offering only predetermined responses set by instructors (Cheng et al. 2024). To more effectively support students, there is a need for more innovative, adaptive mentoring approaches in mathematical problem solving.

ChatGPT has garnered considerable international interest for its role in enhancing student learning (Ngo 2023). As an advanced extension of a large language model (LLM), ChatGPT provides adaptive support by accurately interpreting and responding to user inputs (Chen et al. 2024), offering potential benefits in the swift and efficient acquisition of mathematical concepts and skills (Wardat et al. 2023). However, its occasional instability and unreliable outputs may create confusion and impede students' understanding (Abolkasim and Hasan 2024). Further research is needed to determine how ChatGPT can be integrated into educational frameworks to foster critical thinking without undermining it.

In this regard, this study aims to explore the impact of ChatGPT-facilitated scaffolding on college students' mathematical problem-solving performance, behaviours and perceptions. It will provide valuable insights into the introductory aspects of mathematical instruction at the collegiate level. Furthermore, the findings significantly enhance our understanding of global aspiring teachers' perspectives on the design and implementation of ChatGPT within mathematical education.

2 | Literature Review

2.1 | Mathematical Problem Solving

Problem solving is central to mathematics education, serving both as a key goal and method of learning (National Council of Teachers of Mathematics 2000). It involves understanding the problem, identifying issues, reviewing information, developing strategies, evaluating solutions and taking action to reach a desired outcome (Yuriev et al. 2017). Beyond mathematics, problem solving is essential in daily life, as individuals navigate modern challenges.

Various problem-solving models have been developed to assist students in achieving this learning goal (Branford and Stein 1993; OECD 2018; Sternberg and Sternberg 2006). A well-known model is Sternberg's (Sternberg and Sternberg 2006) seven-step approach, which includes problem identification, problem definition, strategy construction, information organisation, resource allocation and monitoring and evaluation. One strength of this model is its flexibility, allowing for steps to be revisited, reordered or adapted to various problem contexts, including mathematics. However, its generality may limit its effectiveness in more specific domains where targeted approaches are needed. The Programme for International Student Assessment (PISA) 2022 framework offers a more domain-specific model, dividing the problem-solving process into three stages: formulating, employing and interpreting (OECD 2018). The formulating stage involves transforming a situation into a manageable mathematical format, the employing stage focuses on applying mathematical reasoning to find solutions and the interpreting stage involves analysing and evaluating the results and explaining the findings. This model provides a structured approach tailored to mathematical contexts, offering clearer criteria for evaluating student performance.

Despite the utility of these models, undergraduate students continue to face substantial difficulties in mathematical problem solving (Lawson et al. 2020). Studies by Chand et al. (2021) and Joel (2025) report that a significant percentage of students fail mathematics courses in higher education, raising concerns about the adequacy of current instructional practices. One challenge is that mathematical problems often lack fixed solutions, especially in applied contexts where problem solving cannot

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

be reduced to a simple sequence of steps (Hong et al. 2023). Moreover, students frequently encounter challenges during the problem-solving process and limited teaching resources may hinder their ability to solve problems effectively (Høynes et al. 2019). These challenges can undermine students' confidence in learning mathematics, contributing to poor academic performance. To address these issues, there is a growing need for adaptive, responsive forms of support that can scaffold students' learning and promote more effective engagement with mathematical problems.

2.2 | Learning Scaffolding in Mathematical Problem Solving

The concept of learning scaffolding originates from Vygotsky's model of the zone of proximal development (Vygotsky and Cole 1978) and has been widely employed in education to assist students in tackling tasks beyond their current capabilities (Wood et al. 1976). In mathematics education, scaffolding refers to guided assistance provided by a more knowledgeable individual to help students progress towards problem resolution. This support often integrates conceptual knowledge, heuristic strategies and purposeful questioning, delivered in a timely and targeted manner to help learners achieve their goals.

Research has explored the optimal timing, methods and contexts for effective scaffolding (Kim and Hannafin 2011). Technology-enhanced scaffolds are typically classified into four types: conceptual, metacognitive, procedural and strategic (Hill and Hannafin 2001). Conceptual scaffolds help students identify problems and link existing knowledge with new concepts. Metacognitive scaffolds support students in monitoring their thinking and problem-solving processes. Procedural scaffolds assist with navigating the learning environment, while strategic scaffolds encourage exploring alternative solution methods.

The PISA 2022 framework provides a useful lens for aligning scaffolding strategies with specific phases of problem solving (OECD 2018). In the formulating stage, scaffolds support students in organising information, identifying key variables and creating representations. For example, Liang and She (2023) used representational scaffolds to enhance conceptual understanding in online geometry tasks. In the employing stage, scaffolds facilitate idea generation, evaluation of alternatives and evidence-based reasoning. Durandt and Lautenbach (2020) developed strategic scaffolds to boost students' mathematical modelling skills. In the interpreting stage, scaffolds assist in error detection, evaluating reasoning and reflecting on the learning process. Kim and Lim (2019) demonstrated that reflective scaffolding enhances problem-representation skills.

Learning scaffolds that support mathematical problem solving are increasingly diverse and adaptable. Lange et al. (2023) demonstrated that integrating conceptual and strategic scaffolding into a comprehensive instructional model can sustain student interest. Gidalevich and Kramarski (2019) emphasised the importance of gradually reducing scaffolding to support self-regulated learning. Vogel et al. (2022) explored adaptable scaffolding through computer-supported collaborative learning (CSCL) scripts and heuristic worked examples, finding that adaptable CSCL scripts benefited students with advanced self-regulation skills. Notably, most existing approaches emphasise static or semi-dynamic scaffolds, such as pre-designed scripts or fading techniques, with limited attention to real-time, personalised feedback. While some studies have utilised chatbots for mathematical learning (Cheng et al. 2024; Van Doc et al. 2023), these often use rule-based algorithms with fixed responses, limiting their adaptability. In contrast, LLMs, such as AI-based interactive text generators, offer promising potential for delivering dynamic and context-aware scaffolding that enhances personalised support in mathematical problem solving (Jia et al. 2024). However, empirical research on their pedagogical effectiveness and ethical implications is still in its early stages and warrants further investigation.

2.3 | ChatGPT-Supported Scaffolding in Mathematical Problem Solving

Although domain-specific and goal-oriented chatbots have been used to support problem solving (Lee and Yeo 2022), ChatGPT, as an extension of LLMs, offers broader applications in educational settings (Jeon and Lee 2023). It utilises machine learning and natural language processing technologies to interpret and respond to user inputs (OpenAI 2023). Recognised for its improved mathematical capabilities, ChatGPT contributes to educational success by providing foundational knowledge in mathematics and other subjects (Wardat et al. 2023).

Views on ChatGPT's effectiveness in mathematical education are mixed. On one hand, it is acknowledged as a powerful tool that can enhance problem solving and support student learning (Pardos and Bhandari 2024; Wardat et al. 2023). Students can interact with ChatGPT to receive hints and insights for understanding complex mathematical concepts. Moreover, integrating ChatGPT into math education has advantages in raising students' interests (Asare et al. 2023), enriching the learning experience (Gouia-Zarrad and Gunn 2024) and fostering self-regulated learning (Li et al. 2023). On the other hand, technical instabilities, such as occasional irrelevant or incorrect responses, may cause confusion and disrupt learning (Chaka 2023; Rane 2023; Wu 2023). ChatGPT's mathematical reasoning is still limited compared to a graduate-level student, leading to potential inaccuracies (Frieder et al. 2023). Moreover, overreliance on this technology may impair students' critical thinking and problem-solving skills, promoting a superficial understanding of mathematics (Kurban and Şahin 2024; Rane 2023). These concerns underscore the need for strategic implementation and pedagogical guidance when using ChatGPT in mathematics education.

Recent studies have explored how ChatGPT can assist students in mathematical learning. Pardos and Bhandari (2024) demonstrated that ChatGPT-generated hints can produce learning gains comparable to those from human-authored tutoring, suggesting its potential as a scalable instructional tool. Similarly, Wahba et al. (2024) found that integrating ChatGPT into statistics instruction enhanced students' reasoning skills and attitudes towards the subject. These studies primarily focus on learning outcomes rather than the processes through which ChatGPT

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

框架：ChatGPT 鷹架三模組

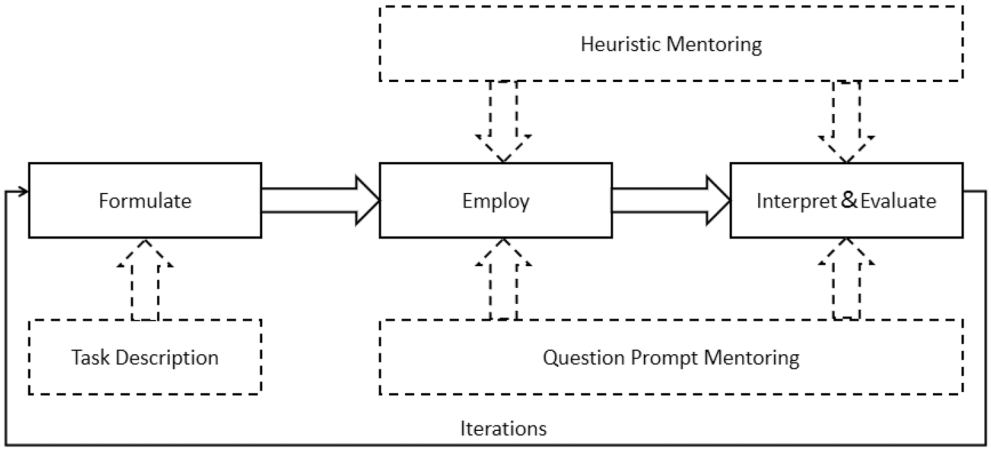


FIGURE 1 | Theoretical framework of ChatGPT-facilitated scaffolding.

supports mathematical thinking. Further research is needed to investigate how ChatGPT can effectively facilitate each stage of the problem-solving process and how its use can be aligned with evidence-based instructional practices.

2.4 | Research Questions

This study aims to investigate the effectiveness of ChatGPT-facilitated scaffolding on college students' mathematical problem-solving performance, behaviours and perceptions. Specifically, this study will address the following research questions:

RQ1. *: To what extent does ChatGPT-facilitated scaffolding enhance students' mathematical problem-solving performances?*

RQ2. *: What distinct behavioural patterns emerge among students when solving mathematical problems with ChatGPT-facilitated scaffolding, and how do these patterns vary between the high-performance group (HPG) and low-performance group (LPG)?*

RQ3. *: What are students' perceptions of their problem-solving practices and their experiences with ChatGPT-facilitated scaffolding?*

3 | Method

3.1 | ChatGPT-Facilitated Scaffolding

To support the stages of mathematical problem solving, this study proposes a scaffolding framework aligned with the PISA 2022 framework (OECD 2018). This alignment ensures theoretical coherence and relevance to international standards for assessing mathematical literacy and problem-solving competencies. The framework incorporates AI-driven interaction grounded in Socratic questioning principles, which are central to fostering higher-order thinking. Socratic questioning encourages individuals to challenge assumptions, uncover underlying truths and engage in evidence-based reasoning to foster cognitive development (Paul and Elder 2016).

In this framework, ChatGPT functions as a dialogic mentor rather than a passive information provider, guiding students through responsive, context-aware interactions. By offering tailored feedback and prompting students to reconsider their reasoning, the AI fosters cognitive construction and cultivates metacognitive awareness. This approach moves beyond conventional answer-based tutoring towards a scaffolded learning model that nurtures independent thinking and problem-solving resilience. As shown in Figure 1, this framework includes three interrelated modules, each targeting specific stages of the problem-solving process:

a. Task description module. This module supports the problem-formulating stage by transforming task descriptions. When students encounter difficulties with a task, they can click on 'Ask AI for help', which automatically directs them to a page where they can engage with ChatGPT to clarify the task.

b. Heuristic mentoring module. Serving as the core of the ChatGPT-facilitated scaffolding, this module supports the employing and interpreting stages through heuristic questioning. ChatGPT acts as a mentor, offering essential cues that encourage students to consider problem-solving strategies rather than providing direct answers.

c. Question prompt mentoring module. This module also supports the employing and interpreting stages by offering question prompts. Specifically designed for students uncertain about what to ask, providing prompts related to 'what', 'why' and 'how'.

The ChatGPT-facilitated scaffolding was integrated into Xiaoya, utilising GPT-4.0 to process messages received through the official OpenAI application programming interfaces (APIs). When students click 'Ask AI for help' (as shown in Figure 2), they are directed to the interface of the scaffolding (Figure 3). This interface includes task description, heuristic mentoring and question prompt modules. Students can select a question from the question prompt module or enter their own solution or inquiry in the text input area.

An AI-powered language model was employed to define the operational parameters of the scaffolding. ChatGPT was

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



FIGURE 2 | Interface of the task.

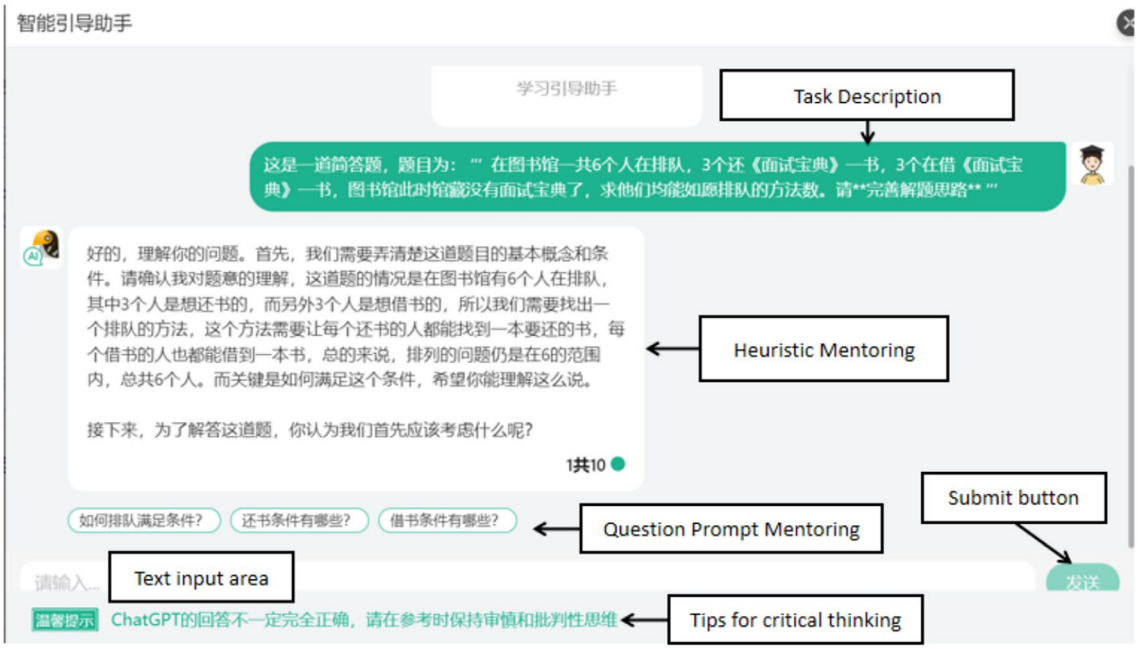


FIGURE 3 | Interface of the ChatGPT-facilitated scaffolding.

TABLE 1 | The settings of ChatGPT-facilitated scaffolding.

Module	Content/Prompts
Task description	The task is as follows: A total of six individuals in the library are lined up, with three returning the book ‘Interview Treasure’ and three borrowing the it. At this time, the library has no more ‘interview Treasure’ in its collection. Please determine the number of possible line-up arrangements.
Heuristic mentoring	As a mathematics teacher, guide students by providing essential cues that encourage independent exploration. Instead of providing direct answers, use questions that prompt students to engage deeply with the problem-solving process.
Question prompt mentoring	After offering heuristic cues, anticipate what students might seek to understand next and generate three related questions to further stimulate inquiry and reflection.

configured to adopt a mentoring role, offering heuristic guidance through guiding questions rather than direct answers and generating follow-up questions aligned with students' inquiries (see Table 1). Specifically, it was programmed to promote student exploration and problem-solving strategies. ChatGPT was assigned the role of a mathematics mentor and instructed to (a) identify underlying mathematical principles, (b) encourage critical thinking through Socratic-style questioning and (c) suggest three contextually relevant follow-up questions for each interaction. Furthermore, the initial task description, including task background and requirements, was pre-determined and uploaded by the instructor to ensure alignment with curriculum standards (see Table 1 for an example task).

AI-student interactions were logged and monitored throughout the study to ensure adherence to the designed scaffolding framework. Researchers periodically reviewed random samples

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of dialogues to evaluate the relevance, appropriateness and heuristic quality of ChatGPT's responses.

3.2 | Research Context

The study was conducted in the 'Combinatorics' course during the autumn of 2023 at a university in central China. This elective course targeted senior students planning to become K-12 teachers. It aimed to impart fundamental combinatorial knowledge and mathematical operation methods. The course adopted a task-driven teaching method, in which students were required to propose methods for solving problems. A key objective of this course was to develop students' abilities to synthesise and apply combinatorial mathematics to problem-solving tasks. Accordingly, students were instructed to use AI to critically analyse and resolve issues, thereby enhancing their analytical capacities.

3.3 | Participants

Twenty-nine third-year undergraduate students (14 females and 15 males) majoring in Mathematics, Applied Mathematics and Computer Science participated in this study. Their participation in this elective course aligned with the specialised training goal of enhancing problem-solving skills. These students had a foundational understanding and some proficiency in using common modern information technology tools. In the initial week, each student completed a Problem Solving Tendency Test. The course instructor randomly assigned students into six groups, each consisting of four to five members, to tackle similar problems in varying contexts.

3.4 | Learning Environment

The study was conducted in a smart classroom, where each student accessed the teaching and learning platform Xiaoya via a mobile phone. Xiaoya (http://www.ai-augmented.com/), independently developed by the research team, offers intelligent educational services to both teachers and students at the university (He et al. 2023; Zhang et al. 2023). It integrates ChatGPT, a generative AI application, to provide problem-solving scaffolding, enabling students to articulate and refine their reasoning. During the introduction to new concepts, students utilised this ChatGPT-facilitated scaffolding to identify and analyse problems (see Figure 3). Engaging in dialogue with the AI, students were guided to understand and critically evaluate the issues, thereby developing solutions. Moreover, the teacher could upload all course materials to Xiaoya, manage coursework and track assignments. In this study, students were required to submit their problem-solving ideas after each task.

3.5 | Experiment Procedure

The course was structured into three phases across eight instructional sessions, each lasting 90 min. The experimental procedure is illustrated in Figure 4.

Phase I introduced foundational topics, including permutations, recurrence relations and generating. Students completed a pre-test

次要：實驗流程

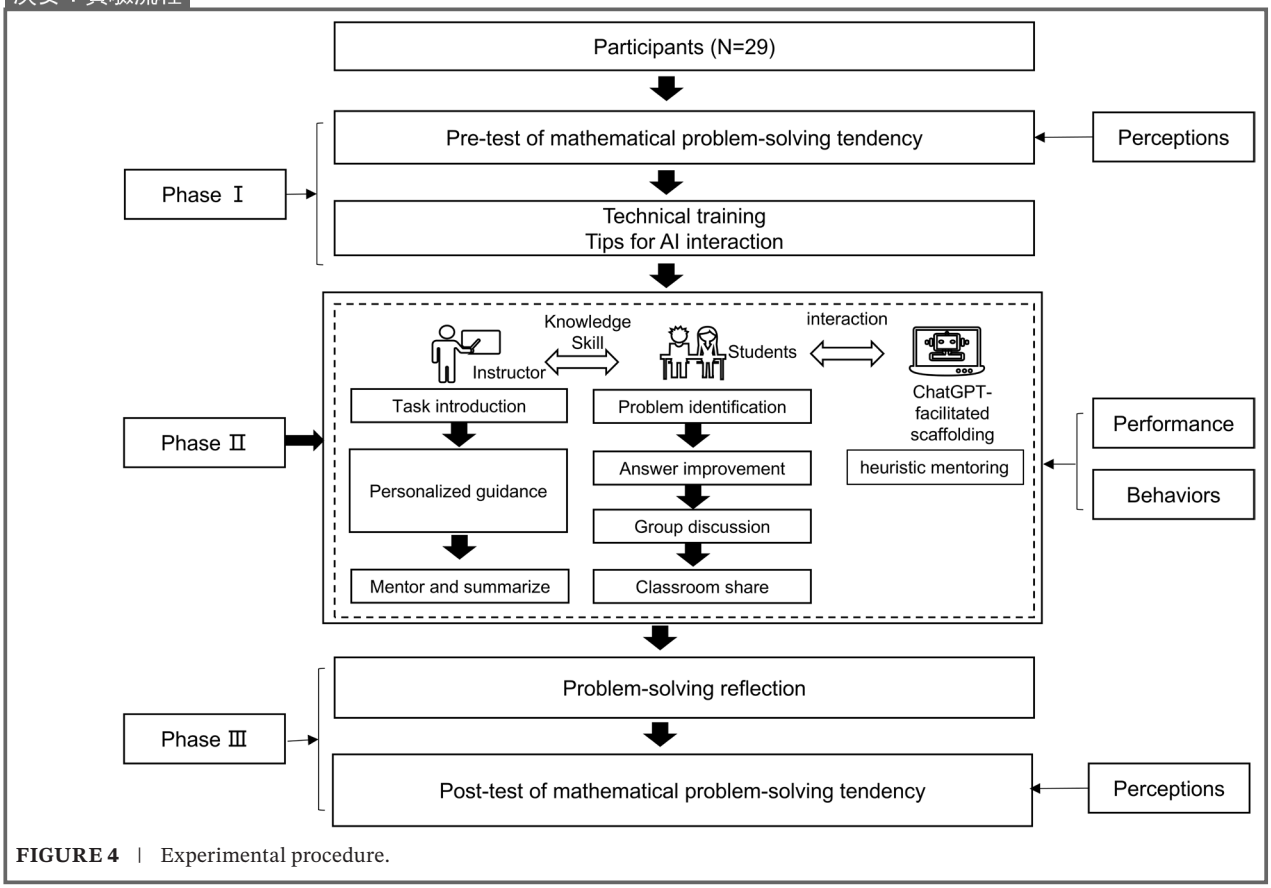


FIGURE 4 | Experimental procedure.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

survey on mathematical problem-solving skills and engaged with ChatGPT-facilitated scaffolding to foster critical communication through guided prompts and iterative dialogue. ChatGPT operated in a controlled mode using customised prompt templates to encourage critical thinking without providing direct answers. Teaching assistants monitored interaction logs to ensure appropriate use and provided support when needed. A question-and-answer session was held to clarify task requirements and students were given time to familiarise themselves with the tools.

Phase II focused on the Catalan number and its calculations. The instructor introduced the tasks, followed by students identifying problems and applying prior knowledge to devise solutions, which they submitted within 5 min. Subsequently, students engaged with the ChatGPT-facilitated scaffolding to refine their responses. During this interaction, ChatGPT operated under a 'Socratic tutor' configuration, designed to prompt students with questions, hints and encouragements rather than explicit answers. This configuration was managed via a pre-programmed dialogue flow to maintain consistency. Students revised and resubmitted their solutions after an additional 5 min of reflection and refinement. Following the AI interaction, students participated in small-group discussions where they shared their experiences with ChatGPT and collaboratively explored strategies to expand and diversify their problem-solving approaches. During these activities, the instructor and teaching assistants provided personalised guidance by circulating among groups, offering feedback and ensuring constructive use of the AI tool. Finally, the groups demonstrated their AI interaction strategies and elaborated on their problem-solving ideas, with the instructor mentoring and summarising the session.

Phase III involved reflection on the problem-solving process. The instructor mentored students on effectively utilising the ChatGPT-facilitated scaffolding to formulate solutions. All participants completed a post-test to assess their mathematical problem-solving perceptions. Throughout the course, the instructor and teaching assistants provided individual support, ensuring a comprehensive educational experience.

3.6 | Instruments

The instruments used in the study included project tasks and surveys on problem-solving skills and emotional experiences, providing metrics on students' performance and perceptions. A coding scheme was implemented to analyse the interactive behaviours with ChatGPT in mathematical problem-solving practices.

3.6.1 | Test Tools

Three similar tasks in different problem contexts, as illustrated in Figure 5, were prepared for students to develop solutions. These tasks included the Theatre Problem, Library Problem and Pancake Problem. Each pair of groups was assigned the same task. During the task exploration phase, students were guided in understanding the Catalan number and its basic calculations.

The tasks were evaluated based on the mathematical framework of PISA (OECD 2018), which categorises mathematical problem solving into three stages (i.e., formulate, employ and interpret and evaluate). The scoring for the formulate and interpret and evaluate stages ranged from 0 to 2, while the employ stage was scored from 0 to 3. The problem-solving evaluation metrics are detailed in Table 2. To ensure consistency, two experts thoroughly discussed and agreed upon the grading criteria for assessing mathematical problem-solving performance. The Spearman's correlation coefficient between the experts' scores was 0.915 ($p < 0.01$). Given this high correlation, the average of their scores was used as the final score for each student.

3.6.2 | Coding Scheme

Mathematical problem-solving stages—formulating, employing and interpreting—were analysed in the context of student interactions with ChatGPT using a question-answer approach that emphasises knowledge construction. Mayer (2002) classified knowledge into factual, conceptual, procedural and metacognitive categories. The first two types refer to 'what' knowledge, while the latter two represent 'how to' knowledge. Explanatory knowledge, which involves understanding principles and logic ('why'), was also proposed (Kim 1994). This study aims to describe problem-solving processes across these domains, specifically examining the construction of knowledge about 'what', 'why', 'how to' and 'what we know' during the employ and interpret and evaluate stages. To ensure the reliability of the coding

Theater Problem:

There are 10 people lined up to enter the theater. The admission fee is \$50, and 5 of the 10 people have only \$50, while the remaining 5 have only \$100. The box office uses an empty cash register to sell tickets for convenience. How many ways are there to line up so that the box office will automatically make change in \$50 bills whenever a person buys a ticket with a \$100 bill?

Library Problem:

A total of 6 people in the library are lined up, with 3 returning the book "Interview Treasure" and 3 borrowing the book "Interview Treasure". At this time, the library has no more "interview Treasure" in its collection. Please find the number of ways they can all get in line as they wish.

Pancake Problem:

There are 16 people in line to buy pancakes, 8 of them have only a \$5 bill each and the other 8 have only a \$10 bill each. Each person buys only one pancake for \$5, and the owner of the pancake store has no change. How many ways can these 16 people arrange themselves so that they don't run out of change?

FIGURE 5 | Three tasks.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Scoring criteria for determining mathematical problem-solving performance.

Score	Formulate	Employ	Interpret and evaluate
0	Wrong interpretation of the question/wrong answers	No method	No justification
1	Incorrect interpretation of questions/did not refer to the requirement of the tasks	Irrelevant problem-solving strategies, resulting in unsuccessful attempts	Justifying answers/ results only
2	Understanding the questions	Relevant/correct problem-solving methods and procedures but incomplete answer/wrong solution	Justifying all problem-solving procedures
3		Relevant/correct problem-solving methods and procedures, leading to the correct answers/result	
	Max score = 2	Max score = 3	Max score = 2

scheme, two content analysis experts reviewed the initial codes representing student-ChatGPT interactions. They then collaborated to finalise the coding scheme, as outlined in Table 3. The last column of the table provides examples translated from Chinese. The notation 'No. 10-P2-1st' refers to the transcription of Student 10's first interaction with ChatGPT during the second problem.

3.6.3 | Questionnaire

To assess students' perceptions of the problem-solving process, a survey instrument initially developed by Heppner and Petersen (2011) was utilised. Sahin et al. (1993) adapted and divided this instrument into six dimensions (i.e., problem-solving confidence, impulsive style, reflective style, avoidant style, monitoring and planfulness), which were originally used to analyse the problem-solving abilities of Turkish students. In this study, the adapted scale was applied to evaluate the differences in students' problem-solving tendencies before and after using ChatGPT-facilitated scaffolding.

To further explore students' perceptions of ChatGPT-facilitated scaffolding, Davis' (1989) Technology Acceptance Model (TAM) was employed as a theoretical framework. This model, which has been widely used to investigate interactions with AI-based learning technologies (Rafique et al. 2020; Sun et al. 2024), posits that perceived usefulness and ease of use are critical factors in the adoption and sustained application of technology (Davis 1989). Additionally, user attitudes and behavioural intentions significantly influence their willingness to integrate technology into learning (Mailizar et al. 2021). Consequently, this study developed a scale within the TAM framework to evaluate students' perceptions of ChatGPT-facilitated scaffolding (additional information can be found in Appendix A). The scale, implemented as a 5-point Likert measure, comprises four subdimensions: perceived usefulness (six items), perceived ease of use (six items), behavioural intention (two items) and attitude (two items). The scale demonstrated strong reliability, with an overall alpha of 0.932 and sub-dimension reliabilities ranging from 0.897 to 0.940. Exploratory factor analysis (EFA) revealed a KMO value of 0.779, with Bartlett's Test of Sphericity confirming the significance ($p = 0.000$), indicating an excellent fit between the model and the data.

3.6.4 | Follow-Up Interviews

To further examine students' experiences with ChatGPT-facilitated scaffolding during the problem-solving process, follow-up semi-structured interviews were conducted with 10 students (five exhibiting higher behavioural intention and five with lower behavioural intention). Each interview lasted approximately 20 min. These interviews provided qualitative insights into how students' problem-solving experiences using ChatGPT-facilitated scaffolding differed from those in traditional collaborative problem-solving contexts (additional details are provided in Appendix B).

3.7 | Data Collection and Analysis

The dataset, which includes performance data, interaction data and self-reported survey data (i.e., questionnaire data and interview data), was analysed using both quantitative and qualitative approaches. The quantitative data was processed using SPSS22.0 to gain insights into students' mathematical problem-solving performance, behaviours and perceptions, employing both descriptive and inferential statistical tests.

Qualitative data, including ChatGPT interaction and interview data, were gathered to address RQ2 and RQ3, respectively. The data, sourced from Xiaoya and video recordings, were coded to explore students' problem-solving processes and their perceptions of the benefits and limitations of ChatGPT-facilitated scaffolding. The interaction data and students' perceptions were systematically organised in Excel and subjected to in-depth content analysis using Nvivo. As outlined in Table 3, the coding scheme was applied to the entire online ChatGPT interaction dataset by the study's first and third authors, using single student utterances as the unit of analysis, consistent with Zhang et al. (2022). To ensure coding reliability, two authors of the

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 | Coding schemes for mathematical problem-solving practices analysis.

Dimensions	Codes	Description	Illustrative example
Formulate			
Problem description	PD	Formulate a mathematical description or a representation that describes a problem.	This is a combinatorial math problem in that a total of six people in the library are lined up, with three returning the book 'Interview Treasure' and three borrowing the book 'Interview Treasure'. At this time, the library has no more 'interview Treasure' in its collection. Please find the number of ways they can all get in line as they wish (No. 10-P2-1st).
Problem contextualization	PC	Decode and make sense of statements, questions or tasks by simplifying or decomposing a situation or a problem to make it amenable to mathematical analysis.	The number of people holding \$50 needs to be greater than the number of people holding \$100 to ensure that change can be made (No. 2-P1-2nd).
Employ			
Prompt selection	PS	Select an appropriate prompt from a list AI provided.	['What is the Catalan number?', 'How is the Catalan number calculated?', 'How do you find $(n+\mathrm{m})!/(n!m!)$?] (No. 25-P3-4th)
Factual representation	FR	Make sense of terminology, specific facts, categories and models to approach the problems of 'what' based on the results of interacting with ChatGPT.	What is dynamic programming (No. 13-P2-6th)?
Procedural representation	PR	Make sense of procedures, techniques and methods to approach the problems of 'how to' based on the results of interacting with ChatGPT.	Now that you have established a recurrence relation, how do you solve for the general term of this recurrence relation (No. 22-P3-15th)?
Interpret and evaluate			
Explanatory representation	ER	Make sense of principles, laws and logic to explain and demonstrate the problems of 'why' based on the results of interacting with ChatGPT.	Why is the solution the fifth Cartland number (No. 5-P1-9th)?
Metacognitive representation	MR	Make sense of strategies to judge and think critically about the problems of 'know what we know' based on the results of interacting with ChatGPT.	You ignored that the people in the queue are different (No. 26-P3-13th)

study had expertise in coding, underwent training and coded them separately. The Cohen's kappa coefficient between the two coders was 0.823, indicating strong agreement. Discrepancies were resolved through discussion and 35 ambiguous or irrelevant discourses were excluded, resulting in 461 coding units from 29 students. These units comprised 98 problem descriptions (PDs), 28 problem contextualisations, 119 prompt selections (PSs), 87 factual representations (FRs), 100 procedural representations (PRs), 14 explanatory representations and 15 metacognitive representations (MRs).

Behavioural patterns are crucial for understanding students' mathematical problem-solving processes and for distinguishing between HPG and LPG. In this study, Lag Sequential Analysis (LSA) was employed to elucidate the sequences of problem-solving behaviours and to assess the statistical significance of specific behaviour patterns. GSEQ 5 software was used to compile the frequency of coded behaviours and the behaviour transition matrix. Adjusted residuals (z scores) were calculated using a z test, with z scores greater than 1.96 indicating statistical significance. A higher adjusted residual value suggests a stronger sequential relationship between variables.

To differentiate between high and low academic performance groups, each student's problem-solving outcomes were considered. Following the methodology from previous studies (Zhang et al. 2022; Zhou et al. 2024), students were classified into three groups based on their problem-solving performance according

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

to the criteria detailed in Table 2. This classification resulted in 8 students being placed in the HPG, 8 in the LPG and the remaining 13 in the IPG (intermediate-performance groups).

4 | Results

4.1 | Impact of ChatGPT-Facilitated Scaffolding on Students' Mathematical Problem-Solving Performance

Regarding RQ1, students' task data was collected and scored based on the criteria outlined in Table 2. The Shapiro–Wilk test, appropriate for assessing normality with small samples, indicated a violation of the normality assumption ($p < 0.05$). Consequently, the Wilcoxon signed-rank test, a non-parametric method, was employed to assess improvements in students' mathematical problem-solving performance following the implementation of ChatGPT-facilitated scaffolding. As depicted in Table 4, significant differences were observed between pre- and post-task scores ($Z = 4.158$, $p = 0.000 < 0.001$), with post-task scores significantly higher. Specifically, significant improvements were noted across all three dimensions, including formulate ($Z = 3.302$, $p = 0.001 < 0.01$), employ ($Z = 4.414$, $p = 0.000 < 0.001$) and interpret & evaluate ($Z = 3.448$, $p = 0.001 < 0.01$). These findings suggest that ChatGPT-facilitated scaffolding may contribute to improvements in students' problem-solving performance. To quantify the Wilcoxon effect size, the Z statistic was divided by the square root of the sample size (Fritz et al. 2011). Most constructs, including problem-solving performance, formulate and interpret and evaluate, exhibited a moderate effect size (0.3–0.5) (Yang and Dalton 2012).

4.2 | Impact of ChatGPT-Facilitated Scaffolding on Students' Mathematical Problem-Solving Behaviours

To address RQ2, students' interaction data during the problem-solving process were collected and coded according to the framework presented in Table 3. Lag sequential analysis was performed to examine the sequence patterns of students' problem-solving behaviours. As summarised in Table 5, a total of 461 behaviours were recorded, with the employ dimension accounting for the largest proportion (66.38%), followed by the formulate dimension (27.33%) and the interpret and evaluate dimension (6.29%). The majority of interactions (93.71%) pertained to the employ and formulate dimensions. Within the formulate dimension, PD was the most frequent, constituting 21.26% of total interactions. In the employ dimension, PS and PRs were the most common, representing 25.81% and 21.69% of interactions, respectively. In the interpret and evaluate dimension, explanatory and MRs were consistently observed, accounting for 3.04% and 3.25% of interactions, respectively.

A comparison of problem-solving processes between HPG and LPG is also provided in Table 5. The HPG exhibited significantly higher percentages in the interpret and evaluate dimension (10.35% vs. 2.17%, $Z = 2.376$, $p < 0.05$) compared to the LPG, while the LPG showed higher percentages in the formulate (26.09% vs. 24.82%) and employ (71.74% vs. 64.83%) dimensions than the HPG. These results suggest that HPG students were more likely to critically explore solutions based on their problem interpretation, whereas LPG students tended to focus on PD and seek solutions through interaction with ChatGPT.

Table 6 presents the transitions, counts and adjusted residuals of significant behaviour sequences (z score > 1.96). A higher adjusted residual indicates a stronger linkage between transitions, with adjusted residuals listed in descending order. A pronounced sequential relationship was observed between PS and PR, with significant transitions noted between these behaviours.

Critical behaviour transitions during the problem-solving process are illustrated in Figure 6, where behaviours in the formulate dimension are marked in yellow, those in the employ dimension in green and those in the interpret and evaluate dimension in pink. The thickness of the arrows represents the proportion of adjusted residual values. Particularly, the most significant behaviour transitions are observed in PS ($f = 6$) and PD ($f = 5$).

Table 7 shows the transitions, counts and adjusted residuals for the students in the HPG. Significant behavioural sequences (z score > 1.96) are highlighted and listed in descending order of their adjusted residuals. The sequential relationship between PS and PR was notably stronger, with these transitions being more significant than other behaviours. Additionally, the transition from FR to MR was particularly pronounced.

Figure 7 provides a visualisation of the behaviour transitions within the HPG, where yellow represents the formulate dimension, green represents the employ dimension and pink represents the interpret and evaluate dimension. The thickness of the arrows corresponds to the strength of the adjusted residual values. It can be seen that PS ($f = 4$) and MR ($f = 4$) had the most significant transition.

結論：解題表現前後測（RQ1）

TABLE 4 | Comparison of problem-solving performance of descriptive statistics and the Wilcoxon signed rank test in the pre-task and post-task.

		Pre-task		Post-task		Wilcoxon signed-rank test			
Variables	**N**	**Mean**	**SD**	**Mean**	**SD**	**Mean rank**	***Z***	**Effect size *R***	***p***
Problem-solving performance	29	2.31	2.316	4.69	2.055	11.50	4.158	0.48	0.000***
Formulate	29	0.90	0.817	1.59	0.568	9.61	3.302	0.44	0.001**
Employ	29	0.86	0.953	1.93	0.799	12.00	4.414	0.52	0.000***
Interpret and evaluate	29	0.55	0.686	1.17	0.848	7.50	3.448	0.37	0.001**

**$p < 0.01$.
***$p < 0.001$.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 | Frequencies of problem-solving behaviour.

Dimensions	Codes	HPG No. (%)	LPG No. (%)	IPG No. (%)	Total No. (%)
Formulate					
Problem description	PD	28 (19.31)	19 (20.65)	51 (22.77)	98 (21.26)
Problem contextualization	PC	8 (5.51)	5 (5.43)	15 (6.70)	28 (6.07)
Sub-total		36 (24.82)	24 (26.09)	66 (29.46)	126 (27.33)
Employ					
Prompt select	PS	38 (26.21)	26 (28.26)	55 (24.55)	119 (25.81)
Factual representation	FR	22 (15.17)	18 (19.57)	47 (20.98)	87 (18.87)
Procedural representation	PR	34 (23.45)	22 (23.91)	44 (19.64)	100 (21.69)
Sub-total		94 (64.83)	66 (71.74)	146 (65.18)	306 (66.38)
Interpret and evaluate					
Explanatory representation	ER	5 (3.45)	2 (2.17)	7 (3.13)	14 (3.04)
Metacognitive representation	MR	10 (6.90)	0 (0)	5 (2.23)	15 (3.25)
Sub-total		15 (10.35)	2 (2.17)	12 (5.36)	29 (6.29)
Total		145 (100)	92 (100)	224 (100)	461 (100)

TABLE 6 | Overall behavioural transition data.

Behavioural transition	Transition count	Adjusted residual
PS → PR	67	10.12
PR → PS	44	5.32
FR → MR	10	5.11
PS → ER	12	4.97
MR → PC	4	3.82
PD → PC	14	3.55
PD → PS	40	3.38
FR → PD	20	2.76
PR → PD	22	2.57
PS → FR	33	2.54
ER → PD	5	2.45
ER → PS	7	2.43

Table 8 shows the transitions, counts and adjusted residuals for the students in LPG. Noteworthy behavioural sequences (z score > 1.96) are similarly highlighted and listed in descending order by their adjusted residuals. The sequential relationship from PS to both FR and PR was stronger, with these transitions standing out compared to other behaviours.

Figure 8 illustrates the behaviour transitions within the LPG, with the same colour coding and arrow thickness representation as in the HPG. The most significant transitions occurred in PS ($f=4$) and PD ($f=4$).

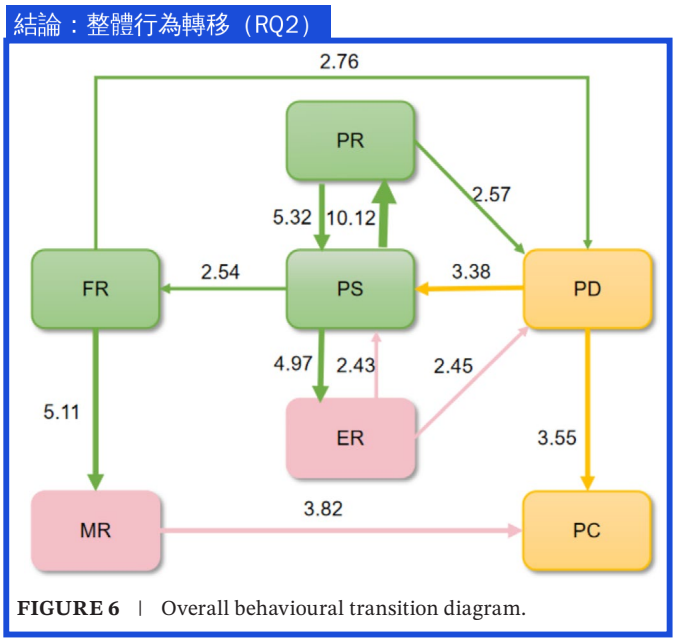


FIGURE 6 | Overall behavioural transition diagram.

4.3 | Undergraduates' Perceptions of Problem-Solving Practice and Experience

Concerning RQ3, student survey responses were collected to investigate their perceptions of the problem-solving process. To validate the uniformity of scores and the normality of data, Levene's test and the Shapiro–Wilk test were conducted, respectively. Levene's test results ($p=0.498$) and dimension values (ranging from 0.162 to 0.928) indicated homogeneity of variance, while Shapiro–Wilk test results ($p=0.095$ in the pre-survey and $p=0.215$ in the post-survey) and dimension values (ranging from 0.076 to 0.800) confirmed normal distribution. These results verified the consistency and normality of the data.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

A paired-sample t test was employed to evaluate changes in students' perceptions of problem-solving practices between the pre-survey and post-survey. As shown in Table 9, there was a significant improvement in students' perceptions ($t = -4.860$, $p < 0.001$). Specifically, notable advancements were observed in impulsive style ($t = -8.104$, $p < 0.001$) and problem-solving confidence ($t = -4.237$, $p < 0.001$). These results suggest that the use of ChatGPT-facilitated scaffolding enhanced students' problem-solving confidence, leading them to approach problems more deliberately by critically evaluating ChatGPT-generated content during their practice.

Student surveys were also used to examine their experiences with ChatGPT-facilitated scaffolding after the experiment. Descriptive results (see Table 10) indicated that most construct scores were above 3.5, with students perceiving ChatGPT as both useful for solving problems and convenient to use. Specifically, in terms of perceived usefulness, students believed ChatGPT could help generate effective strategies (mean = 3.96; SD = 0.73) and promote problem-solving efficacy (mean = 3.93; SD = 0.65). Regarding ease of use, students found it convenient to use (mean = 3.80; SD = 0.71) and reported proficiency in using the tool (mean = 3.81; SD = 0.81). Overall, students had a positive attitude towards ChatGPT-facilitated scaffolding (mean = 3.63; SD = 0.98), enjoying their interactions and expressing satisfaction with the solutions provided. However, scores related to intention to use were relatively lower than other dimensions (mean = 3.48; SD = 0.97).

To further investigate students' experiences with ChatGPT-facilitated scaffolding, follow-up interviews were conducted with 10 students (five with higher behavioural intention and five with lower behavioural intention). Most students acknowledged the value of ChatGPT in providing personalised support tailored to their learning needs. One student remarked, 'It's impressive how the scaffolding adapts and modifies the solutions provided based on the prompt'. Additionally, ChatGPT offered an interactive, mentor-like experience, guiding students to think critically rather than providing direct answers. One student noted, 'It doesn't give a direct answer but encourages you to think through the problem'. Another commented on the helpfulness of ChatGPT's prompt scaffolding, 'It gives prompts, which is useful when you're unsure of what to ask'.

However, students with lower intentions to use the scaffolding reported challenges. Some encountered irrelevant or excessive responses, which deviated from the required knowledge points. These lengthy replies were often overwhelming and hindered students' ability to focus on their learning goals. One student explained, 'The lengthy responses often didn't meet my needs and required more critical thinking, leading to cognitive overload'.

TABLE 7 | Behavioural transition data in HPG.

Behavioural transition	Transition count	Adjusted residual
PS → PR	24	6.44
FR → MR	7	5.33
PR → PS	17	3.50
MR → FR	4	2.69
PS → ER	4	2.66
ER → PD	2	2.58
MR → PC	2	2.38
MR → PS	12	2.00

次要：低表現組行為轉移

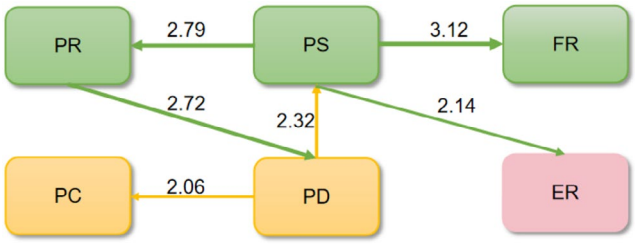


FIGURE 8 | Behavioural transition diagram of LPG.

TABLE 8 | Behavioural transition data in LPG.

Behavioural transition	Transition count	Adjusted residual
PS → FR	11	3.12
PS → PR	12	2.79
PR → PD	6	2.72
PD → PS	10	2.32
PS → ER	2	2.14
PD → PC	3	2.06

次要：高表現組行為轉移

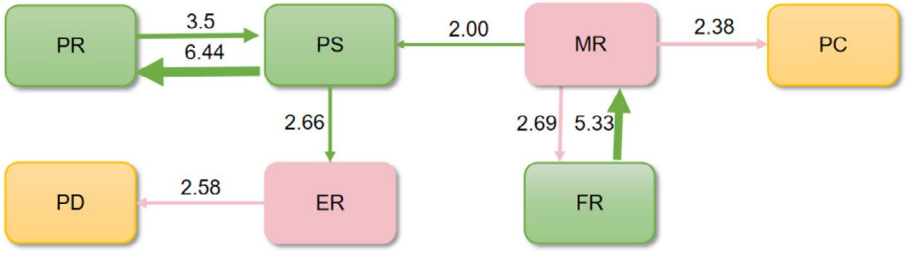


FIGURE 7 | Behavioural transition diagram of HPG.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

結論：學生知覺前後測（RQ3）

TABLE 9 | The paired sample *t* test on undergraduates' perceptions of problem-solving practice.

	Paired differences					
			95% CI of the difference			
Pairs	**Mean**	**SD**	**Lower**	**Upper**	***t***	**Probability (two tailed)**
Pre-test and Post-test	−0.408	0.436	−0.580	−0.235	−4.860	0.000
Formulate						
Planfulness	−0.074	0.554	−0.293	0.145	−0.695	0.493
Employ						
Impulsive style	−1.376	0.882	−1.725	−1.027	−8.104	0.000
Reflective style	−0.111	0.575	−0.338	−0.116	−1.005	0.324
Avoidant style	−0.213	0.685	−0.484	−0.058	−1.615	0.118
Interpret and evaluate						
Monitorin	−0.247	0.761	−0.548	0.054	−1.685	0.104
Problem-solving confidence	−0.426	0.523	−0.633	−0.219	−4.237	0.000

TABLE 10 | Descriptive results on undergraduates' perceptions of experiences with ChatGPT-facilitated scaffolding.

Dimensions of the survey		**Mean**	**SD**
Perceived usefulness	Learning efficacy	3.93	0.65
	Strategy generation	3.96	0.73
	Knowledge mastery	3.81	0.75
Perceived ease of use	Manoeuvrability	3.80	0.71
	Interactivity	3.72	0.86
	Proficiency	3.81	0.81
Intention to use		3.48	0.97
Attitude		3.63	0.98

5 | Discussion

This study examined the potential impact of ChatGPT-facilitated scaffolding on college students' mathematical problem solving by comparing pre-task and post-task performance, analysing problem-solving patterns and assessing students' perceptions.

First, the findings indicated a statistically significant improvement in students' problem-solving performance after the intervention. Consistent with Patero (2023), integrating ChatGPT into mathematics instruction improved learning outcomes. Improvements were particularly notable across the formulating, employing and evaluating stages. The scaffolding enabled students to decompose tasks into sub-problems, supporting learning within their zone of proximal development and promoting the integration of prior and new knowledge (Kusmaryono et al. 2021; Puntambekar 2022). Heuristic mentoring provided intuitive cues for reasoning, while targeted questions encouraged critical thinking and deeper exploration (Barak 2013; Koichu et al. 2007; Li 2023). However, the absence of a control group limits the ability to attribute these improvements solely to the scaffolding intervention.

Second, students showed varied use of the scaffolding. Frequency analysis revealed that most interactions occurred during the employing stage, particularly through PS and PR. This supports the findings by Song and Becker (2014), who noted that novices benefit from decomposing problems and linking them to prior knowledge. Additionally, differences emerged between the HPG and LPG in the evaluating stages. HPG students more critically examined solutions based on their problem interpretation, consistent with prior research linking metacognitive awareness to improved performance (Akben 2020; Fleur et al. 2021; Muncer et al. 2022). Although AI scaffolding may facilitate reflective thinking through heuristic questioning, additional metacognitive interventions are needed. Integrating features that identify metacognitive states—such as monitoring interaction frequency, tone and word choice—could better trigger metacognitive-guided interventions. Behavioural transition patterns further underscored group differences in the use of scaffolding. A prominent transition was observed from PS to PR, suggesting that the prompts played a key role in guiding students' thinking. This aligns with Kauffman et al. (2008), who found that prompts support goal clarification and self-monitoring. HPG students exhibited a notable transition from factual to MR, reflecting deeper engagement. In contrast, LPG students tended to transition from PS to FR, relying more on acquiring basic knowledge through procedural methods. This mirrors findings by Zhou et al. (2024), who reported that LPG students often remain at surface-level engagement, limiting their problem-solving development. While these patterns are informative, the small sample size may limit the generalisability of the findings. Future research could explore the effects of adding more explanatory and reflective prompts to improve problem-solving outcomes.

Third, students expressed generally positive perceptions of ChatGPT-facilitated scaffolding. Survey results indicated

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

increased confidence and reduced impulsive problem-solving styles, suggesting a shift towards more deliberate and reflective thinking. This finding aligns with Lee et al. (2024), who emphasised the value of student-led solution development for critical thinking. Similarly, increased self-confidence aligns with Chang and Hwang's (2024) research, which demonstrates that students gain confidence when verifying AI-generated information independently. While most students responded positively to the scaffolding, intention to use it in the future received slightly lower ratings. Follow-up interviews provided further insight into both the benefits and limitations of the intervention. Students highlighted personalised support and heuristic mentoring, consistent with the broader literature on AI-facilitated learning (e.g., Sun et al. 2024). Reported benefits included round-the-clock accessibility, natural language interaction, multilingual capabilities and clear conceptual explanations. However, several drawbacks were also identified, including occasional irrelevant or overly lengthy replies and technical issues. These concerns echo those raised in prior studies (e.g., Yilmaz and Yilmaz 2023), which caution against over-reliance on AI tools due to the potential for information overload, reduced motivation and user frustration. Future iterations should focus on enhancing the relevance and brevity of responses and integrating mechanisms to detect and respond to learners' metacognitive states to better support self-regulated learning.

6 | Conclusion

This study introduced ChatGPT-facilitated scaffolding and explored its potential to support undergraduates' mathematical problem-solving performances, behaviours and perceptions. The scaffolding comprised three modules, including task description, heuristic mentoring and question prompt mentoring. Findings from this mixed-methods study indicate that the scaffolding may positively influence students' academic performance and problem-solving confidence, with most participants reporting favourable perceptions of the scaffolding. The study demonstrates the feasibility of integrating generative AI tools, such as ChatGPT, into mathematics curricula. Observed problem-solving patterns also offer preliminary insights that may inform the design of future educational interventions aimed at developing students' problem-solving skills.

Despite these promising findings, the study has limitations. First, the absence of a control group restricts a comprehensive understanding of the impact of ChatGPT-facilitated scaffolding. Although an integrated analysis approach was employed to strengthen the findings, including a control group would provide a more rigorous analysis of the causal relationship between scaffolding and problem-solving outcomes. Second, the study was conducted within a single university mathematics course, where factors such as teacher and peer involvement and the reliability of ChatGPT, may have influenced students' performance. Additionally, the short duration of the experiment and the small sample size of 29 students may have limited the identification of broader problem-solving patterns. Future research should address these limitations by incorporating a control group, expanding the sample size and extending the experiment duration. A design-based research approach could be employed to refine the scaffolding further and examine the new design principles on a larger scale of participants, thereby enhancing the validity and generalisability of the findings.

Author Contributions

Ruijie Zhou: conceptualization, methodology, investigation, data curation, formal analysis, visualization, writing – original draft. **Xiuling He:** conceptualization, supervision, funding acquisition, validation. **Qiong Fan:** conceptualization, methodology, writing – review and editing, supervision, project administration, resources. **Yangyang Li:** writing – review and editing, supervision, data curation. **Yue Li:** supervision, writing – review and editing. **Xiong Xiao:** data curation, writing – review and editing. **Jing Fang:** writing – review and editing, supervision.

Ethics Statement

All procedures performed in studies involving human participants were in accordance with ethical standards. The study was approved by the Social Sciences and Humanities Research Ethics Committee of Central China Normal University.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

Abolkasim, E., and M. Hasan. 2024. "Integrating ChatGPT in Education and Learning: A Case Study on Libyan Universities." *Journal of Pure and Applied Sciences* 23, no. 2: 19–24. https://doi.org/10.51984/jopas.v23i2.3082.

Ahmadi Safa, M., and F. Motaghi. 2024. "Cognitive vs. Metacognitive Scaffolding Strategies and EFL Learners' Listening Comprehension Development." *Language Teaching Research* 28, no. 3: 987–1010. https://doi.org/10.1177/13621688211021821.

Akben, N. 2020. "Effects of the Problem-Posing Approach on Students' Problem Solving Skills and Metacognitive Awareness in Science Education." *Research in Science Education* 50, no. 3: 1143–1165. https://doi.org/10.1007/s11165-018-9726-7.

Álvarez, J. A. M., E. G. Arnold, E. A. Burroughs, E. W. Fulton, and A. Kercher. 2020. "The Design of Tasks That Address Applications to Teaching Secondary Mathematics for Use in Undergraduate Mathematics Courses." *Journal of Mathematical Behavior* 60: 100814. https://doi.org/10.1016/j.jmathb.2020.100814.

Asare, B., Y. D. Arthur, and F. O. Boateng. 2023. "Exploring the Impact of ChatGPT on Mathematics Performance: The Influential Role of Student Interest." *Education Science and Management* 1, no. 3: 158–168. https://doi.org/10.56578/esm010304.

Barak, M. 2013. "Impacts of Learning Inventive Problem-Solving Principles: Students' Transition From Systematic Searching to Heuristic Problem Solving." *Instructional Science* 41, no. 4: 657–679. https://doi.org/10.1007/s11251-012-9250-5.

Blum, W., and M. Niss. 1991. "Applied Mathematical Problem Solving, Modelling, Applications, and Links to Other Subjects—State, Trends and Issues in Mathematics Instruction." *Educational Studies in Mathematics* 22, no. 1: 37–68. https://doi.org/10.1007/BF00302716.

Branford, J. D., and B. S. Stein. 1993. *The Ideal Problem Solver: A Guide for Improving Thinking, Learning and Creativity*. WH Freeman.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chaka, C. 2023. "Generative AI Chatbots - ChatGPT Versus YouChat Versus Chatsonic: Use Cases of Selected Areas of Applied English Language Studies." *International Journal of Learning, Teaching and Educational Research* 22, no. 6: 1–19. http://ijlter.myres.net/index.php/ijlter/article/view/1638.

Chand, S., K. Chaudhary, A. Prasad, and V. Chand. 2021. "Perceived Causes of Students' Poor Performance in Mathematics: A Case Study at Ba and Tavua Secondary Schools." *Frontiers in Applied Mathematics and Statistics* 7: 614408. https://doi.org/10.3389/fams.2021.614408.

Chang, C.-C., and G.-J. Hwang. 2024. "ChatGPT-Facilitated Professional Development: Evidence From Professional Trainers' Learning Achievements, Self-Worth, and Self-Confidence." *Interactive Learning Environments* 33, no. 1: 883–900. https://doi.org/10.1080/10494820.2024.2362798.

Chen, J., Z. Liu, X. Huang, et al. 2024. "When Large Language Models Meet Personalization: Perspectives of Challenges and Opportunities." *World Wide Web* 27, no. 4: 42. https://doi.org/10.1007/s11280-024-01276-1.

Cheng, L., E. Croteau, S. Baral, C. Heffernan, and N. Heffernan. 2024. "Facilitating Student Learning With a Chatbot in an Online Math Learning Platform." *Journal of Educational Computing Research* 62, no. 4: 907–937. https://doi.org/10.1177/07356331241226592.

Csanadi, A., I. Kollar, and F. Fischer. 2021. "Pre-Service Teachers' Evidence-Based Reasoning During Pedagogical Problem-Solving: Better Together?" *European Journal of Psychology of Education* 36, no. 1: 147–168. https://doi.org/10.1007/s10212-020-00467-4.

Davis, F. D. 1989. "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology." *MIS Quarterly* 13, no. 3: 319. https://doi.org/10.2307/249008.

Durandt, R., and G. Lautenbach. 2020. "Strategic Support to Students' Competency Development in the Mathematical Modelling Process: A Qualitative Study." *Perspectives in Education* 38, no. 1: 211–223. https://doi.org/10.18820/2519593X/pie.v38i1.15.

Fleur, D. S., B. Bredeweg, and W. van den Bos. 2021. "Metacognition: Ideas and Insights From Neuro- and Educational Sciences." *npj Science of Learning* 6, no. 1: 1–11. https://doi.org/10.1038/s41539-021-00089-5.

Frieder, S., L. Pinchetti, R.-R. Griffiths, et al. 2023. "Mathematical Capabilities of ChatGPT." *Advances in Neural Information Processing Systems* 36: 27699–27744. https://proceedings.neurips.cc/paper_files/paper/2023/hash/58168e8a92994655d6da3939e7cc0918-Abstract-Datasets_and_Benchmarks.html.

Fritz, C., P. Morris, and J. Richler. 2011. "Effect Size Estimates: Current Use, Calculations, and Interpretation." *Journal of Experimental Psychology. General* 141: 2–18. https://doi.org/10.1037/a0024338.

Gabriel, F., S. Buckley, and A. Barthakur. 2020. "The Impact of Mathematics Anxiety on Self-Regulated Learning and Mathematical Literacy." *Australian Journal of Education* 64, no. 3: 227–242. https://doi.org/10.1177/0004944120947881.

Gidalevich, S., and B. Kramarski. 2019. "The Value of Fixed Versus Faded Self-Regulatory Scaffolds on Fourth Graders' Mathematical Problem Solving." *Instructional Science* 47, no. 1: 39–68. https://doi.org/10.1007/s11251-018-9475-z.

Gouia-Zarrad, R., and C. Gunn. 2024. "Enhancing Students' Learning Experience in Mathematics Class Through ChatGPT. International Electronic Journal of." *Mathematics Education* 19, no. 3: em0781. https://doi.org/10.29333/iejme/14614.

He, X., J. Fang, H. N. H. Cheng, Q. Men, and Y. Li. 2023. "Investigating Online Learners' Knowledge Structure Patterns by Concept Maps: A Clustering Analysis Approach." *Education and Information Technologies* 28, no. 9: 11401–11422. https://doi.org/10.1007/s10639-023-11633-8.

Heppner, P. P., and C. H. Petersen. 2011. "Problem-Solving Inventory [Dataset]." https://doi.org/10.1037/t04336-000.

Hill, J. R., and M. J. Hannafin. 2001. "Teaching and Learning in Digital Environments: The Resurgence of Resource-Based Learning." *Educational Technology Research and Development* 49, no. 3: 37–52. https://doi.org/10.1007/BF02504914.

Hong, W., J. R. Star, R.-D. Liu, R. Jiang, and X. Fu. 2023. "A Systematic Review of Mathematical Flexibility: Concepts, Measurements, and Related Research." *Educational Psychology Review* 35, no. 4: 104. https://doi.org/10.1007/s10648-023-09825-2.

Høynes, S.-M., T. Klemp, and V. Nilssen. 2019. "Mentoring Prospective Mathematics Teachers as Conductors of Whole Class Dialogues – Using Video as a Tool." *Teaching and Teacher Education* 77: 287–298. https://doi.org/10.1016/j.tate.2018.10.014.

Jeon, J., and S. Lee. 2023. "Large Language Models in Education: A Focus on the Complementary Relationship Between Human Teachers and ChatGPT." *Education and Information Technologies* 28, no. 12: 15873–15892. https://doi.org/10.1007/s10639-023-11834-1.

Jia, J., T. Wang, Y. Zhang, and G. Wang. 2024. "The Comparison of General Tips for Mathematical Problem Solving Generated by Generative AI With Those Generated by Human Teachers." *Asia Pacific Journal of Education* 44, no. 1: 8–28. https://doi.org/10.1080/02188791.2023.2286920.

Joel, L. O. 2025. "Students' Failure in Mathematics: A Case Study of Calculus-Related Modules at a University in Johannesburg." *International Journal of Innovative Technology and Interdisciplinary Sciences* 8, no. 1: 1. https://doi.org/10.15157/IJITIS.2025.8.1.294-312.

Jonte, M., B. Witzel, S. Powell, et al. 2022. "A Meta-Analysis of Mathematics Word-Problem Solving Interventions for Elementary Students Who Evidence Mathematics Difficulties." *Review of Educational Research* 92: 003465432110700. https://doi.org/10.3102/00346543211070049.

Kauffman, D. F., X. Ge, K. Xie, and C.-H. Chen. 2008. "Prompting in Web-Based Environments: Supporting Self-Monitoring and Problem Solving Skills in College Students." *Journal of Educational Computing Research* 38, no. 2: 115–137. https://doi.org/10.2190/EC.38.2.a.

Kim, J. 1994. "Explanatory Knowledge and Metaphysical Dependence." *Philosophical Issues* 5: 51–69. https://doi.org/10.2307/1522873.

Kim, J. Y., and K. Y. Lim. 2019. "Promoting Learning in Online, Ill-Structured Problem Solving: The Effects of Scaffolding Type and Metacognition Level." *Computers & Education* 138: 116–129. https://doi.org/10.1016/j.compedu.2019.05.001.

Kim, M. C., and M. J. Hannafin. 2011. "Scaffolding Problem Solving in Technology-Enhanced Learning Environments (TELEs): Bridging Research and Theory With Practice." *Computers & Education* 56, no. 2: 403–417. https://doi.org/10.1016/j.compedu.2010.08.024.

Koichu, B., A. Berman, and M. Moore. 2007. "The Effect of Promoting Heuristic Literacy on the Mathematical Aptitude of Middle-School Students." *International Journal of Mathematical Education in Science and Technology* 38, no. 1: 1–17. https://doi.org/10.1080/00207390600861161.

Kurban, C. F., and M. Şahin. 2024. "Educational Implications." In *The Impact of ChatGPT on Higher Education*, 153–179. Emerald Publishing Limited. https://doi.org/10.1108/978-1-83797-647-820241009.

Kusmaryono, I., Jupriyanto, and W. Kusumaningsih. 2021. "Construction of Students' Mathematical Knowledge in the Zone of Proximal Development and Zone of Potential Construction." *European Journal of Educational Research* 10, no. 1: 341–351. https://eric.ed.gov/?id=EJ1284001.

Lange, C., A. Gorbunova, E. Shmeleva, and J. Costley. 2023. "The Relationship Between Instructional Scaffolding Strategies and Maintained Situational Interest." *Interactive Learning Environments* 31, no. 10: 6640–6651. https://doi.org/10.1080/10494820.2022.2042314.

Lawson, D., M. Grove, and T. Croft. 2020. "The Evolution of Mathematics Support: A Literature Review." *International Journal of Mathematical*

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Education in Science and Technology 51, no. 8: 1224–1254. https://doi.org/10.1080/0020739X.2019.1662120.

Lee, D., and S. Yeo. 2022. "Developing an AI-Based Chatbot for Practicing Responsive Teaching in Mathematics." *Computers & Education* 191: 104646. https://doi.org/10.1016/j.compedu.2022.104646.

Lee, H.-Y., P.-H. Chen, W.-S. Wang, Y.-M. Huang, and T.-T. Wu. 2024. "Empowering ChatGPT With Guidance Mechanism in Blended Learning: Effect of Self-Regulated Learning, Higher-Order Thinking Skills, and Knowledge Construction." *International Journal of Educational Technology in Higher Education* 21, no. 1: 16. https://doi.org/10.1186/s41239-024-00447-4.

Li, H. 2023. "Effects of a ChatGPT-Based Flipped Learning Guiding Approach on Learners' Courseware Project Performances and Perceptions." *Australasian Journal of Educational Technology* 39, no. 5: 5. https://doi.org/10.14742/ajet.8923.

Li, P.-H., H.-Y. Lee, Y.-P. Cheng, A. I. Starčič, and Y.-M. Huang. 2023. "Solving the Self-Regulated Learning Problem: Exploring the Performance of ChatGPT in Mathematics." In *Innovative Technologies and Learning*, edited by Y.-M. Huang and T. Rocha, 77–86. Springer Nature. https://doi.org/10.1007/978-3-031-40113-8_8.

Liang, C.-P., and H.-C. She. 2023. "Investigate the Effectiveness of Single and Multiple Representational Scaffolds on Mathematics Problem Solving: Evidence From Eye Movements." *Interactive Learning Environments* 31, no. 6: 3882–3897. https://doi.org/10.1080/10494820.2021.1943692.

Lu, D., and Y.-N. Xie. 2024. "The Application of Educational Technology to Develop Problem-Solving Skills: A Systematic Review." *Thinking Skills and Creativity* 51: 101454. https://doi.org/10.1016/j.tsc.2023.101454.

Mailizar, M., D. Burg, and S. Maulina. 2021. "Examining University Students' Behavioural Intention to Use e-Learning During the COVID-19 Pandemic: An Extended TAM Model." *Education and Information Technologies* 26, no. 6: 7057–7077. https://doi.org/10.1007/s10639-021-10557-5.

Marco, N., and A. Palatnik. 2024. "Teachers Pose and Design Context-Based Mathematics Tasks: What Can Be Learned From Product Evolution?" *Educational Studies in Mathematics* 115, no. 2: 223–246. https://doi.org/10.1007/s10649-023-10271-8.

Mayer, R. E. 2002. "A Taxonomy for Computer-Based Assessment of Problem Solving." *Computers in Human Behavior* 18, no. 6: 623–632. https://doi.org/10.1016/S0747-5632(02)00020-1.

Muncer, G., P. A. Higham, C. J. Gosling, S. Cortese, H. Wood-Downie, and J. A. Hadwin. 2022. "A Meta-Analysis Investigating the Association Between Metacognition and Math Performance in Adolescence." *Educational Psychology Review* 34, no. 1: 301–334. https://doi.org/10.1007/s10648-021-09620-x.

National Council of Teachers of Mathematics. 2000. *Principles and Standards for School Mathematics*. Reston, VA: Author.

National Research Council, & Mathematics Learning Study Committee. 2001. *Adding It Up: Helping Children Learn Mathematics*. National Academies Press.

Ngo, T. T. A. 2023. "The Perception by University Students of the Use of ChatGPT in Education." *International Journal of Emerging Technologies in Learning (iJET)* 18, no. 17: 4–19. https://doi.org/10.3991/ijet.v18i17.39019.

OECD. 2018. "PISA 2022 Mathematics Framework (Draft)." https://pisa2022-maths.oecd.org/files/PISA%202022%20Mathematics%20Framework%20Draft.pdf.

OpenAI. 2023. "ChatGPT: Optimizing Language Models for Dialogue." https://openai.com/blog/chatgpt/.

Ouyang, F., Z. Chen, M. Cheng, Z. Tang, and C.-Y. Su. 2021. "Exploring the Effect of Three Scaffoldings on the Collaborative Problem-Solving Processes in China's Higher Education." *International Journal of Educational Technology in Higher Education* 18, no. 1: 35. https://doi.org/10.1186/s41239-021-00273-y.

Pardos, Z. A., and S. Bhandari. 2024. "ChatGPT-Generated Help Produces Learning Gains Equivalent to Human Tutor-Authored Help on Mathematics Skills." *PLoS One* 19, no. 5: e0304013. https://doi.org/10.1371/journal.pone.0304013.

Patero, J. L. 2023. "Revolutionizing Math Education: Harnessing ChatGPT for Student Success." *International Journal of Advanced Research in Science, Communication and Technology* 3, no. 1: 807–813. https://doi.org/10.48175/IJARSCT-12375.

Paul, R., and L. Elder. 2016. *The Thinker's Guide to Socratic Questioning*. Rowman & Littlefield. https://sc.panda985.com/extdomains/books.google.com/books/about/The_Thinker_s_Guide_to_Socratic_Question.html?hl=zh-CN&id=ADWbDwAAQBAJ.

Puntambekar, S. 2022. "Distributed Scaffolding: Scaffolding Students in Classroom Environments." *Educational Psychology Review* 34, no. 1: 451–472. https://doi.org/10.1007/s10648-021-09636-3.

Rafique, H., A. O. Almagrabi, A. Shamim, F. Anwar, and A. K. Bashir. 2020. "Investigating the Acceptance of Mobile Library Applications With an Extended Technology Acceptance Model (TAM)." *Computers & Education* 145: 103732. https://doi.org/10.1016/j.compedu.2019.103732.

Rane, N. 2023. "Enhancing Mathematical Capabilities Through ChatGPT and Similar Generative Artificial Intelligence: Roles and Challenges in Solving Mathematical Problems." *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4603237.

Säfström, A. I., J. Lithner, T. Palm, et al. 2024. "Developing a Diagnostic Framework for Primary and Secondary Students' Reasoning Difficulties During Mathematical Problem Solving." *Educational Studies in Mathematics* 115, no. 2: 125–149. https://doi.org/10.1007/s10649-023-10278-1.

Sahin, N., N. H. Sahin, and P. P. Heppner. 1993. "Psychometric Properties of the Problem Solving Inventory in a Group of Turkish University Students." *Cognitive Therapy and Research* 17, no. 4: 379–396. https://doi.org/10.1007/BF01177661.

Song, T., and K. Becker. 2014. *Expert vs Novice: Problem Decomposition/Recomposition in Engineering Design. 2014 International Conference on Interactive Collaborative Learning (ICL)*, 181–190. IEEE. https://doi.org/10.1109/ICL.2014.7017768.

Sternberg, R. J., and K. Sternberg. 2006. *Cognitive Psychology (Second Edition)*. Thomson/Wadsworth.

Sun, D., A. Boudouaia, C. Zhu, and Y. Li. 2024. "Would ChatGPT-Facilitated Programming Mode Impact College Students' Programming Behaviors, Performances, and Perceptions? An Empirical Study." *International Journal of Educational Technology in Higher Education* 21, no. 1: 14. https://doi.org/10.1186/s41239-024-00446-5.

Van Doc, N., N. T. Hoai Nam, N. T. Thanh, and N. Minh Giam. 2023. "Teaching Mathematics With the Assistance of an AI Chatbot to Enhance Mathematical Thinking Skills for High School Students." *International Journal of Current Science Research and Review* 6, no. 12: 8574–8580. https://doi.org/10.47191/ijcsrr/V6-i12-102.

Vogel, F., I. Kollar, F. Fischer, K. Reiss, and S. Ufer. 2022. "Adaptable Scaffolding of Mathematical Argumentation Skills: The Role of Self-Regulation When Scaffolded With CSCL Scripts and Heuristic Worked Examples." *International Journal of Computer-Supported Collaborative Learning* 17, no. 1: 39–64. https://doi.org/10.1007/s11412-022-09363-z.

Vygotsky, L. S., and M. Cole. 1978. *Mind in Society: Development of Higher Psychological Processes*. Harvard University Press. https://books.google.com.hk/books?hl=zh-CN&lr=&id=RxjjUefze_oC&oi=fnd&pg=PA1&dq=Mind+in+Society&ots=okwUP1s-fs&sig=g--9d3qgchUM_JrXrlBN7W23GfM&redir_esc=y#v=onepage&q=Mind%20in%20Society&f=false.

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Wahba, F., A. O. Ajlouni, and M. A. Abumosa. 2024. "The Impact of ChatGPT-Based Learning Statistics on Undergraduates' Statistical Reasoning and Attitudes Toward Statistics. Eurasia Journal of Mathematics, Science and Technology." *Education* 20, no. 7: em2468. https://doi.org/10.29333/ejmste/14726.

Wardat, Y., M. A. Tashtoush, R. AlAli, and A. M. Jarrah. 2023. "ChatGPT: A Revolutionary Tool for Teaching and Learning Mathematics. Eurasia Journal of Mathematics, Science and Technology." *Education* 19, no. 7: em2286. https://doi.org/10.29333/ejmste/13272.

Wood, D., J. S. Bruner, and G. Ross. 1976. "The Role of Tutoring in Problem Solving." *Journal of Child Psychology and Psychiatry* 17, no. 2: 89–100. https://doi.org/10.1111/j.1469-7610.1976.tb00381.x.

Wu, Y. 2023. "Integrating Generative AI in Education: How ChatGPT Brings Challenges for Future Learning and Teaching." *Journal of Advocacy, Research and Education* 2, no. 4: 6–10. https://doi.org/10.56397/JARE.2023.07.02.

Yang, D., and J. E. Dalton. 2012. *A Unified Approach to Measuring the Effect Size Between Two Groups Using SAS*. In SAS global forum. vol. 335, 1–6. SAS Institute. https://support.sas.com/resources/papers/proceedings12/335-2012.pdf.

Yilmaz, R., and F. G. K. Yilmaz. 2023. "The Effect of Generative Artificial Intelligence (AI)-Based Tool Use on Students' Computational Thinking Skills, Programming Self-Efficacy and Motivation." *Computers and Education: Artificial Intelligence* 4: 100147. https://doi.org/10.1016/j.caeai.2023.100147.

Yuriev, E., S. Naidu, L. S. Schembri, and J. L. Short 2017. "Scaffolding the Development of Problem-Solving Skills in Chemistry: Guiding Novice Students out of Dead Ends and False Starts." *Chemistry Education Research and Practice* 18, no. 3: 486–504. https://doi.org/10.1039/C7RP00009J.

Zhang, S., Q. Gao, M. Sun, et al. 2022. "Understanding Student Teachers' Collaborative Problem Solving: Insights From an Epistemic Network Analysis (ENA)." *Computers & Education* 183: 104485. https://doi.org/10.1016/j.compedu.2022.104485.

Zhang, S., H. Li, Y. Wen, Y. Zhang, T. Guo, and X. He. 2023. "Exploration of a Group Assessment Model to Foster Student Teachers' Critical Thinking." *Thinking Skills and Creativity* 47: 101239. https://doi.org/10.1016/j.tsc.2023.101239.

Zhou, R., Y. Li, X. He, C. Jiang, J. Fang, and Y. Li. 2024. "Understanding Undergraduates' Computational Thinking Processes: Evidence From an Integrated Analysis of Discourse in Pair Programming." *Education and Information Technologies* 29: 19367–19399. https://doi.org/10.1007/s10639-024-12597-z.

Appendix A

Survey for Student Perception Towards ChatGPT-Facilitated Learning Scaffolding

Perceived usefulness

Using ChatGPT-facilitated learning scaffolding during problem-solving practices helps me complete tasks faster.

Using ChatGPT-facilitated learning scaffolding improves my problem-solving performance.

Using ChatGPT-facilitated learning scaffolding helps me generate more problem-solving strategies.

Using ChatGPT-facilitated learning scaffolding during my tasks can improve my problem-solving efficiency.

I strengthened my understanding during the ChatGPT-facilitated problem-solving process.

ChatGPT-facilitated learning scaffolding is advantageous for my task completion.

Perceived ease of use

It is easy for me to learn to use ChatGPT-facilitated learning scaffolding.

I will find it easy to get the materials that I want from ChatGPT.

My interactions with ChatGPT are clear and easy to understand.

I find that ChatGPT can be flexible in its interactions.

It is easy for me to become proficient with ChatGPT.

I find ChatGPT-facilitated learning scaffolding easy to operate and use.

Intention to use

Teachers/instructors should make more use of ChatGPT-facilitated learning scaffolding in classroom instruction to guide learning.

I hope to apply ChatGPT-facilitated learning scaffolding to math problem solving on an ongoing basis.

Attitude

I enjoy interacting with ChatGPT.

I have a strong sense of satisfaction after getting the right problem-solving idea in interacting with ChatGPT.

Appendix B

Interview protocol

Do you like using ChatGPT-facilitated learning scaffolding to help you solve mathematical problems and why?

What services does this scaffolding offer you?

How do you feel using this scaffolding in this course so far?

How do you use this scaffolding during the problem-solving process?

What do you think is good about this scaffolding?

What are the biggest problems you've encountered with this scaffolding and what strategies do you adapt to resolve them?

13652729, 2025, 4, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/jcal.70077 by National Taiwan Normal University, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License